

Question	Answer	Marks
2(a)	product of mass and velocity	B1
2(b)(i)	$\text{maximum speed} = 7.2 \times 10^4 / 1800$ $= 40 \text{ m s}^{-1}$	A1
2(b)(ii)	$\text{kinetic energy} = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 1800 \times 40^2$ $= 1.4 \times 10^6 \text{ J}$	C1
		A1
2(b)(iii)	$a = \text{gradient of line} / \text{mass}$ or $a = (v - u) / t$ or $a = \Delta v / t$	C1
	$a = \text{e.g. } (3.6 \times 10^4) / (4.0 \times 1800) = 5.0 \text{ m s}^{-2}$ or $a = \text{e.g. } 20 (-0) / 4 = 5.0 \text{ m s}^{-2}$ or $F = \text{e.g. } 3.6 \times 10^4 / 4.0 = 9.0 \times 10^3$ and $a = 9.0 \times 10^3 / 1800 = 5.0 \text{ m s}^{-2}$	A1
2(b)(iv)	$\text{distance} = \frac{1}{2} \times 40 \times (8 + 4)$ or $\text{distance} = \frac{1}{2} \times 72\,000 \times (8 + 4) / 1800$ or $\text{distance} = [40^2 / (2 \times 5)] + [40^2 / (2 \times 10)]$ or $\text{distance} = \frac{1}{2} \times 5 \times 8^2 + (40 \times 4 - \frac{1}{2} \times 10 \times 4^2)$	C1
	distance = 240 m	A1

Question	Answer	Marks
2(c)	stepped shape, showing one constant value up to 8.0 s then stepping to a different constant value from 8.0 s with no time delay	B1
	horizontal straight line from (0, 5.0) to (8.0, 5.0)	B1
	horizontal straight line from (8.0, –10.0) to (12.0, –10.0)	B1

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